AMSAT Linear Transponder Module (LTM) Interface Control Document

January 31, 2026
Revision: V2.5

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ICD Revision History

Revision	Date	Author(s)	Change Log
0.1	February 28, 2021	E. Skoog (K1TVV) B. Fisher (WB1FJ)	ICD initial draft material
0.2	May 16, 2021	B. Fisher (WB1FJ)	More CAN and downlink details
0.3	June 14, 2021	B. Fisher (WB1FJ)	Combine .2 with UMO's appendix revisions; update telemetry to maximum length
0.4	June 15, 2021	B. Fisher (WB1FJ)	Fix frame size and max CAN sizes
0.5	July 27, 2021	B. Fisher (WB1FJ)	Fix field order in CAN ID; clarify MAX31725 addresses
0.6	August 19, 2021	B. Fisher (WB1FJ)	Add CAN health message types
0.7	September 9, 2021	B. Fisher (WB1FJ)	Add CAN uplink messages
0.8	December 8, 2021	B. Fisher (WB1FJ)	Fix typos, remove host appendix
0.9	January 19, 2022	E.Skoog (K1TVV)	Change ITAR to proprietary
1.0	January 20, 2022	B. Fisher (WB1FJ)	Change telem bits from 1 per data bytes to 8 per data byte. Clear up some text.
1.1	January 21, 2022	B. Fisher (WB1FJ)	Change 32-bit binary downlink to 3 CAN messages

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1.2	January 22, 2022	B. Fisher (WB1FJ)	Change 32-bit binary downlink to up to 4 CAN messages
2.0	September 12, 2024	B. Fisher (WB1FJ)	Delete references to host-specific appendix and provide only a single set of data connections.
2.1	September 14, 2024	B. Fisher (WB1FJ)	Clean up entire document, including sections numbering.
2.2	September 15, 2024	B. Fisher (WB1FJ)	Fixes from Mike Moore review comments and other updates.
2.3	September 21, 2024	B. Fisher (WB1FJ) E. Skoog (K1TVV)	Fixes from Eric, tweaks from Burns
2.4	November 19, 2025	M. Moore (K4MVM) B. Fisher (WB1FJ)	Change CAN speed, and other updates
2.5	January 31, 2026	B. Fisher (WB1FJ)	Change in telemetry to work with the actual implementation

1. INTRODUCTION

Interface management and control is a System Engineering activity that begins in parallel with the development of system architectures and continues for the life of a system. Likewise, the evolutionary process to define interfaces begins with concept definition and continues throughout the development phase. Interfaces, both internal and external to the system, are documented and formally managed. This activity is intended to ensure compatibility among subsystems being designed and fabricated, and to ensure future interoperability between systems.

System interfaces are both physical and functional. Initially, a system is depicted by a top-level Concept Diagram that identifies its external interfacing elements. Source documents, such as Capability Development Documents, Capstone Requirements Documents, system specifications, and interface documents provide interface requirements to ensure interoperability between systems and that the required capabilities are realized. An operational concept may also provide descriptions, interactions, and requirements between the system and its external elements. An interface definition process aids architecture and requirements evolution in conjunction with the overall system definition process, i.e., interfaces will mature as operational and system requirements mature.

1.1. Overview

An interface, from the perspective of system development, can be identified as any point where a system and something, or someone, meet. This interface point may include other systems, internal hardware, external peripherals, networks, system users, etc.

Interfaces are documented using Interface Control Documents (ICD) that describe inter/intra system workings including any communication rules. ICDs help ensure compatibility between system segments and components and are considered critical elements of effective system design and development.

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Interface Control Document (ICD) ≡ describes the interworking of two elements of a system that share a common interface. An ICD may also describe the interaction between a user and a system, a software component and a hardware device, or two software or hardware components. An ICD is typically used where complex interfaces exist between components that are being developed by different teams.

For the development team, ICDs define applicable system inputs, outputs, and interfaces to other systems, subsystems, and/or users. An ICD should only describe the interface itself and not the characteristics of the systems connected. An ICD does not necessarily include details regarding the meaning or intended use of the data being interchanged across the interface. Any design features, functions, or logic supporting a system interface should be detailed in separate design and/or specification documents. Defining an ICD in this way allows other teams to develop connecting systems without being concerned about design details or how data is treated by the other system. This allows development teams to work without knowledge of the business logic or technical aspects behind the interfacing system. This facilitates modularity and leads to easier maintenance and extension of the interfacing systems.

1.2. Purpose

The purpose of interface management and control is to define interface requirements so as to ensure compatibility between inter-connected pieces of equipment and to provide an authoritative means of controlling the design of the interfaces.

The purpose of an ICD is to clearly communicate a system's external inputs and outputs for all potential interface actions whether or not they are transparent to the system's users. An ICD may describe the system interface to its lowest physical or functional elements (e.g., units, modules, software routines/tasks, etc.) and/or parameter values (e.g., circuits, voltages, impedances).

An ICD describes the relationship between two components of a system in terms of data items and messages passed, protocols used, and the timing and sequencing of interactive events. Specifically, ICDs are generated to:

- Ensure configuration control of the system's interface design to prevent any changes to characteristics that would affect its compatibility with other systems,
- Define and illustrate the physical and functional characteristics of a system in sufficient detail to ensure compatibility of the interface, and verification of that compatibility can be determined from the information in the ICD,
- Facilitate the identification of missing interface data and control the submission of that data, and
- Communicate coordinated design decisions and design changes to all involved participants.

ICD's by nature are requirements documents. They define design requirements and allow integration. They record the agreed-to design solution to interface requirements and provide a control mechanism to ensure that the agreed-to designs are not changed by one participant without the knowledge of and agreement with the other participant(s).

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1.3. Interface Requirements

It is important that requirement statements defining the interface between two systems be stated in clear, unambiguous language. Starting with the type and purpose of the interface, all interface design requirements and constraints must be identified. Explicit definitions of the content and format of every message/file that can pass between the systems and the conditions under which each is to be sent should be detailed. For non-IT interfaces, assembly characteristics, stresses, connections, accessibility requirements, mechanical details/descriptions, etc. should be provided.

Tables, diagrams, and figures should be used to aid in understanding interface characteristics. Reference material such as data dictionaries, protocols, and standards should be identified in an ICD References section.

Interface requirements may be summarized by describing data protocols, communication methods and processing priorities used by the interface, e.g.,

- Data protocols may include messages and ASCII files exchanged,
- Communication methods may include networks or storage media, and
- If specific information processing must be prioritized, formatted and communicated in accordance with schedules (real time – batch – asynchronous – etc.), such information should be clearly highlighted.

1.4. Cooperative Agreement

This Interface Control Document specifies the interface between AMSAT's Linear Transponder Module (LTM) and a Partner University's CubeSat Platform (the host). The information contained herein is to be used during development and testing by all parties requiring such information, including hardware/software developers, system engineers, and testers responsible for verifying the required performance of the interface. Upon approval by both development organizations, this ICD shall be incorporated into the requirements baseline of each system.

This ICD is intended to be identical for all Partner Universities (also called Partners below), and as such is not intended to be changed during the development of the host system, unless serious problems are discovered by either party. There will be no unilateral/unpublished changes made to the interface details/requirements included in this ICD. Rather, any mission or design changes that would affect the interface information contained herein will be immediately brought to the attention of the other party so that impact analyses and possible redirection planning may be conducted.

To ensure effective/efficient development of the capabilities detailed in this ICD, Partner University and AMSAT engineering team members will coordinate/collaborate technical/programmatic issues via scheduled and unscheduled communication utilizing documentation generation/exchange/review/revision activities, email, telephone, video-teleconference, network-enabled test events, etc.

Mutual support will be provided to assist in trouble-shooting during initial module (engineering model – EM) testing, Flight Model (FM) integration testing, Preliminary and Critical Design Reviews (PDR/CDR), Mission Readiness Reviews (MRR), and other launch preparation activities as required. In particular, the Partner University shall report to AMSAT any issues where the LTM behavior does not appear to be consistent with this ICD. AMSAT may ask the Partner University to perform certain tests which are not possible for AMSAT to perform before the LTM is integrated with the host.

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2. SYSTEM INTERFACE DESCRIPTIONS

This ICD specifies the interfaces between AMSAT's LTM and a Partner University CubeSat host platform. It is divided into three main sections: (1) System Description; (2) Structure and Electrical Lines; and (3) Software and Data, to include protocols and uses for the electrical connections.

The terms "will", "shall", or "must" herein mean that the specified action (or lack of in the case of "not") is a requirement placed on either the LTM or the Partner University's CubeSat design. The term "may" means that the specified action is allowed but optional.

The term "opaque to LTM" is sometimes used in these descriptions. LTM does not act on opaque data, but will downlink, save, or pass it to the host platform depending on the context. The term "ignored by LTM" means that LTM does not act on the data, nor does it save or transmit it.

In several places, the ICD specifies optional LTM support for host devices (such as temperature sensors). The host may choose not to use these devices, in which case an invalid value may be downlinked in telemetry.

2.1 AMSAT Linear Transponder Module (LTM) Hardware and Signals

System Description
The LTM is a 3-board module provided by AMSAT comprising: an LIHU (LTM Internal Housekeeping Unit, essentially the LTM's on-board computer), an ICR (Improved Command Receiver), and an RxTx (the linear transponder receiver – transmitter and telemetry transmitter). It is intended to be used as the radio module for a non-AMSAT developed satellite, herein called "the host platform." All software for the LIHU is provided by AMSAT.

The LIHU is NOT intended to be used as a flight computer for the host platform. It communicates with the host platform's flight computer(s) via a CAN bus, and optionally with some other host devices by I2c or single-purpose lines (described below).

AMSAT's LTM will provide VHF uplink command and control capability as well as UHF downlink telemetry data. The downlink telemetry consists of a 1200bps BPSK modulated channel which may contain LTM and Host operating parameters and operational status, as well as educational, and scientific data from the host's subsystems. The design of the transponder includes state-of-the-art integrated circuit components to provide exceptional communications performance, while minimizing size, weight and power. For Amateur Radio Satellite Service authorized use, the LTM will provide a V/u transponder (re-transmitter). Amateur radio satellite uplink transmissions to be repeated will require generation of VHF (2 meter) SSB or CW amateur radio signals; the repeated downlink signal receipt will require reception of UHF (70 cm) SSB/CW signals.

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2.1.1 Block Diagram

At the conceptual level, AMSAT's Linear Transponder Module can be envisioned as a set of major functional blocks as illustrated in Figure 1 below.

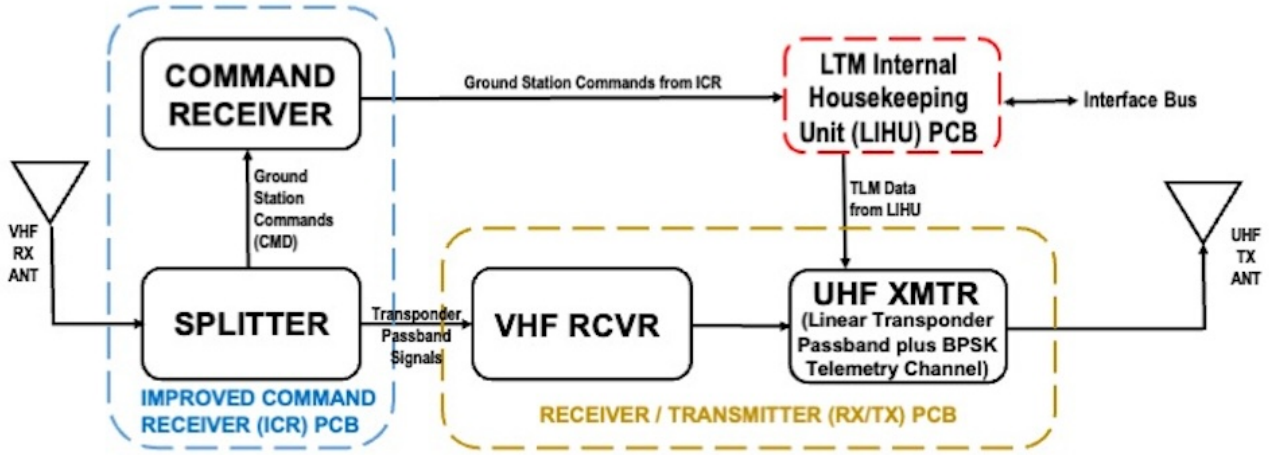


Figure 1: AMSAT Linear Transponder Module (LTM) Top Level Block Diagram

Figure 2 below illustrates the major interface connections between AMSAT's Linear Transponder Module (LTM) and a University's CubeSat host platform**.**

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AMSAT Linear Transponder Module Interface Diagram

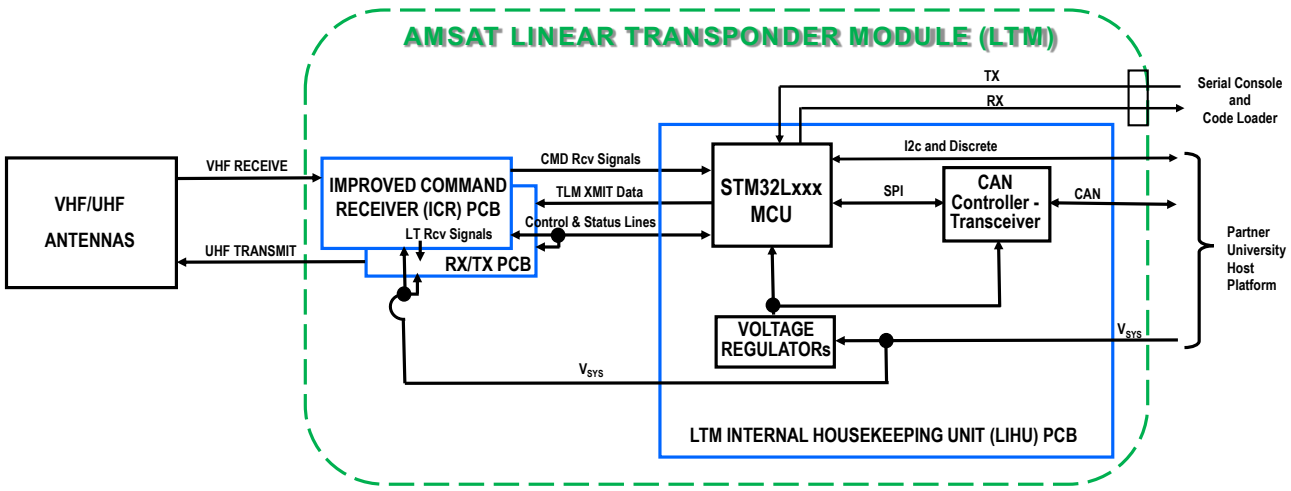


Figure 2: Linear Transponder Module Major Interface Connections

2.1.2 RF Frequencies

The LTM transponder 3dB bandwidth is 30 kHz. All uplink-downlink VHF and UHF operational frequencies to be used by AMSAT's LTM shall be coordinated and licensed through IARU and FCC by the university partner, viz.,

UPLINK: 145.xxx MHz to 145.xxx MHz

DOWNLINK: 43X.xxx MHz to 43X.xxx MHz (inverting)

TLM Downlink Channel (dedicated): 435.xxx MHz (1200 bps BPSK)

COMMAND & CONTROL UPLINK CHANNEL: *[redacted]* MHz (exact frequency is sensitive)

2.1.3 RF Performance Parameters

The set of performance parameters for the four main Linear Transponder Module (LTM) RF signal chains: UHF Passband Transmit (TX); (2) VHF Passband Receive (RX); (3) VHF Uplink Command Channel Receive; and (4) the UHF Downlink Telemetry Transmit Channel. are shown in Tables 1 to 4 below, viz.:

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PERFORMANCE PARAMETERS	REQUIRED VALUES	COMMENTS
UHF Downlink Transmit (TX) Frequency Range	TBD to TBD MHz	VHF Passband receive frequencies inverted
Transponder Bandwidth	30 kHz (3 dB)	
Maximum RF Output Power	1W (+30 dBm) max.	Linear Transponder Power Amplifier $P_{out\ peak}$ (SSB full loading; including TLM channel)
Frequency Stability (L.O.)	±25 ppm	-20° C to +50° C
Spurious Products	≥ -20 dBc	
Component Temp. Range	-40° C to +85° C	
DC Power (maximum)	~2.6 W (625 mA @ 5.2V)	Transmitter power is dynamic and depends on transponder loading and Average Power Tracking (APT) performance

Table 1: UHF Passband Transmit (TX) Signal Chain Performance Values

PERFORMANCE PARAMETERS	REQUIRED VALUES	COMMENTS
VHF Uplink (Transponder) Receive Frequency Range	TBD to TBD MHz	
Transponder Bandwidth	30 kHz (3 dB)	
Minimum Discernable Signal (MDS)	≈ -110 dBm	for 10dB S/N SSB signals
Frequency Stability (L.O.)	±25 ppm	-20° C to +50° C
Noise Figure (NF)	<3.0 dB	
Third Order Output Intercept Point (OIP3)	>5 dBm	
IMD Levels	>-50 dBc	
Component Temp. Range	-40 °C to +85° C	
DC Power (Maximum)	~208 mW (40 mA @5.2V)	Includes Command Channel Receiver

Table 2: VHF Passband Receive (RX) Signal Chain Performance Values

PERFORMANCE PARAMETERS	REQUIRED VALUES	COMMENTS
Telecommand Uplink Receive Channel Frequency	TBD MHz	Exact frequency is sensitive information
Channel Bandwidth	≈ 20 kHz	NBFM (± 5kHz Deviation)
Minimum Discernable Signal (MDS)	-110 dBm for a BER of 1 x 10^{-4}	AFSK-FM (see Command and Control Interface Specification) – SENSITIVE DESIGN INFORMATION
Noise Figure (NF)	≤ 3.0 dB	
Component Temp. Range	-40° C to +85° C	

Table 3: VHF Uplink Command Channel Receive Signal Chain Performance Values

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PERFORMANCE PARAMETERS	REQUIRED VALUES	COMMENTS
UHF Telemetry Downlink Transmit (TX) Frequency	TBD MHz	**1200 bps (channel rate) BPSK**
Telemetry Downlink Channel Maximum Output Power	≈100 mW (+20 dBm) max.	**Peak Power**
Frequency Stability (L.O.)	±25 ppm	**-20° C to +50° C**
Component Temp. Range	-40° C to +85° C	
TLM TX ONLY; DC Power (maximum)	1.1 W (210 mA @ 5.2 V)	**@100 mW RF telemetry-beacon RF output**

Table 4: UHF Downlink Telemetry Transmit Signal Chain Performance Values

2.1.4 LTM Structure

The LTM is not designed to pass PC-104 connector bus lines through from one end of the stack to the other. Therefore, the LTM must be located at either end of a Partner University's PC-104-compatible PCB stack. A Partner University's interfacing bus connector can be a PC-104 compatible male or female mate. The LTM mechanical design consists of three Printed Circuit Boards (PCB) built to the dimensions shown in Figure 3 below. The LTM's PCBs are 2 and 4-layer FR-4 material with ~8 mm inter-board spacing and interconnected by SAMTEC QTH/QSH connectors, a TX/RX - ICR inter-board RF coax, and a PC-104 bus connector. Figure 3 below also shows a 3D rendering of the three board LTM.

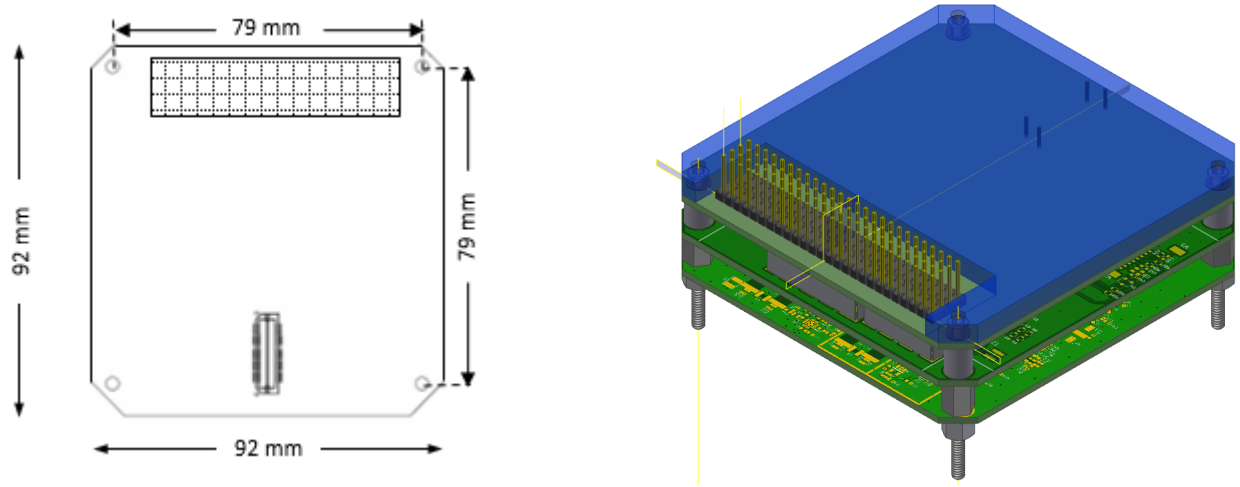


Figure 3: LTM PCB Dimensions and 3D Rendering

Refer to Figure 4 below. This shows the position of the PC104 relative to the mounting holes on the board. It is better, however, to use the precise connector layout dimensions and location as can be measured on the LTM's LIHU Gerbers in the KiCad files.

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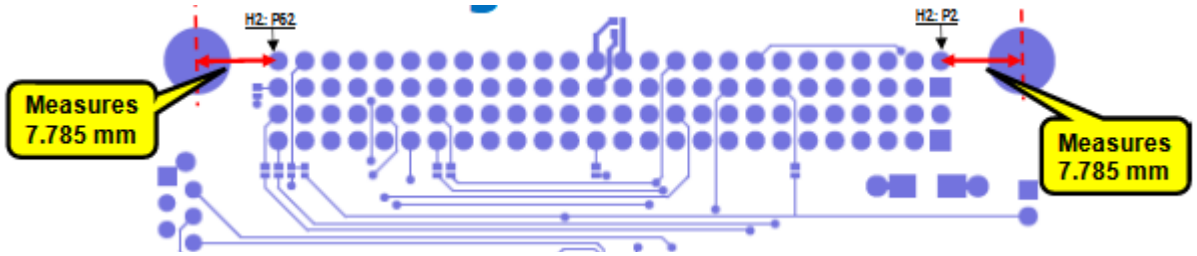


Figure 4: LTM PCB PC104 Location Relative to Mounting Holes

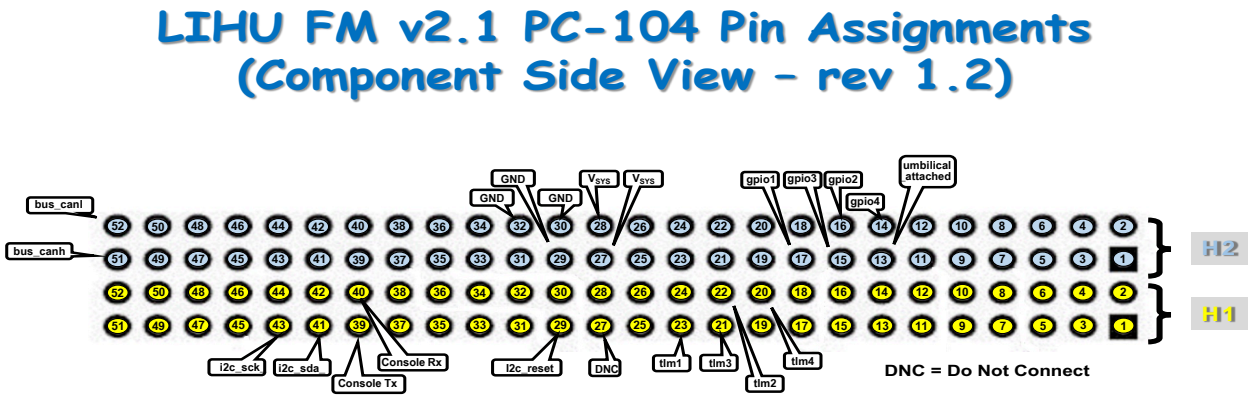


Figure 5: LTM PC-104 pin assignments (component side view)

2.1.5 Electrical Signal Connections

The major signal interface lines between the LTM and the Partner University CubeSat subsystems are provided via a PC-104 connector (see Figure 5). This connector can be provided as either male or female and can be arranged so the LTM can be either on the bottom or the top of the host platform's board stack. The University Partner must specify the gender and end as an early part of its communication with AMSAT. The LTM receives primary DC power over this connector via two Vsys and two GND pins. The Vsys lines are equivalent, as are the GND pins i.e., they are connected together in parallel and supplied by the host platform's Electrical Power Subsystem (EPS).

2.2 Test and Operational Configurations

The LTM interface functions are designed to provide reliable VHF uplink and UHF downlink operations once the Partner University's CubeSat host platform is in orbit.

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However, before this, during development, all required interface functions should be verified through RF and baseband testing via connections between (laboratory simulated) ground station command data and the Partner University's CubeSat platform. To support initial Partner University Lab development/testing, AMSAT will provide an early LTM EM unit along with a Familiarization and Acceptance Test Procedure (ATP) document. This will allow the Partner University's Development Team to get familiar with the LTM design and operation via a laboratory test configuration set-up as illustrated in Figure 6 below:

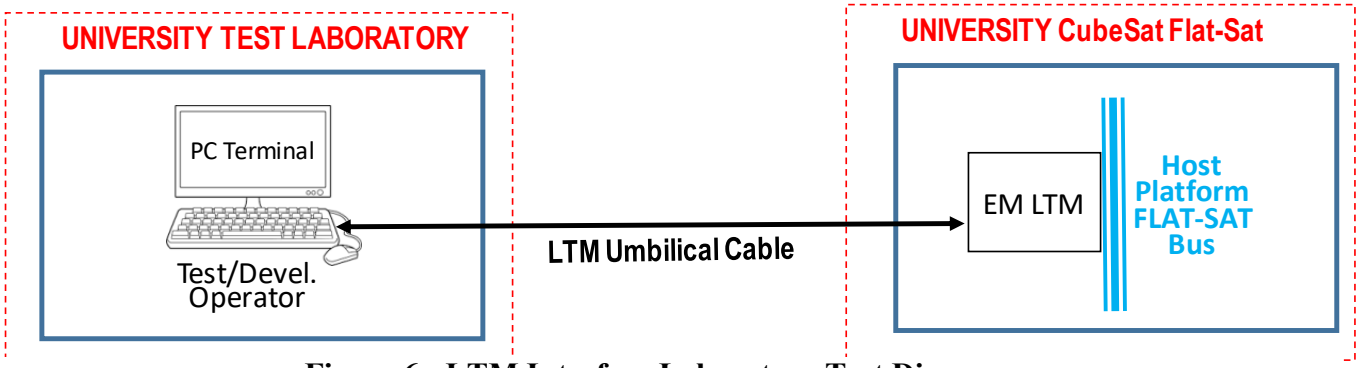


Figure 6: LTM Interface Laboratory Test Diagram

While on orbit, LTM supported ground communications with a University CubeSat should be provided through the operational configuration illustrated in Figure 7 below:

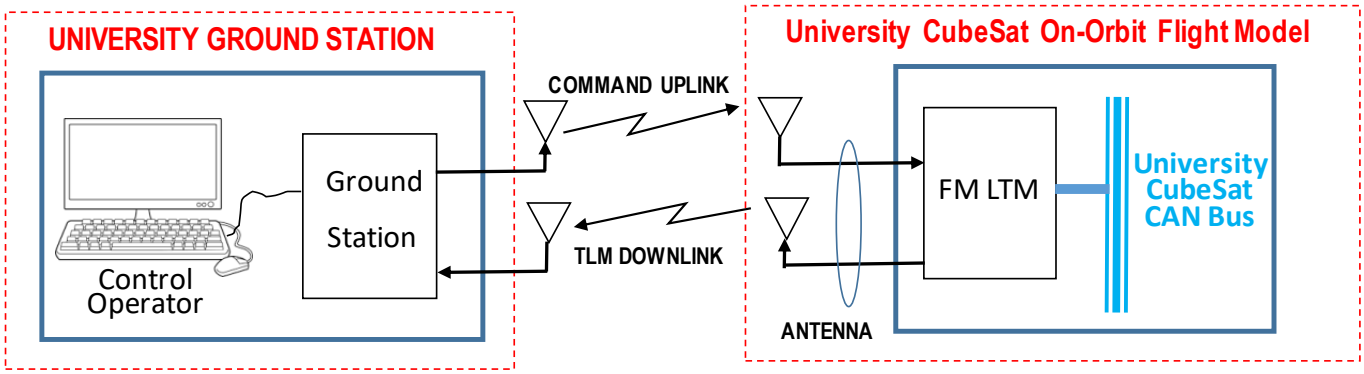


Figure 7: On-Orbit LTM - University CubeSat Communications Diagram

Additionally, LTM subsystem functions will support amateur radio satellite VHF/UHF transponder operations. These functions will be controlled by AMSAT provided ground station application. The AMSAT application for commanding the CubeSat is called AMCOM (AMSAT COMmand); it will be provided along with AMCOM authorization files for university-specified licensed amateur radio operators. AMSAT's ground station application is called FoxTelem. These applications employ some of the same VHF uplink and UHF downlink capabilities utilized by the host platform's other subsystems.

2.3 Software Behavior

2.3.1 The Attached Signal – When the LTM is being tested on the ground, we expect the "Umbilical Attached" signal (H2 Pin 13) to be raised (digital high). This will prevent the transmitter from switching on and will also boot faster and behave slightly differently in case of a failure. In some AMSAT satellites, "Attached" also prevents release of any deployables. We recommend that the host use a similar model, but we require that however it is done, no transmissions from the LTM are allowed without a 50 ohm dummy load or a 50 ohm fully deployed antenna attached.

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2.3.2 "Console" or Terminal Commands

When the LTM is on the ground, users should attach a serial terminal operating at 56kbaud to connector H1, pins 39 and 40. This terminal will allow users to set up the LTM, give test commands, and request printed information from it. AMSAT's usual procedure is to ensure that when a terminal is installed, Umbilical Attached is automatically raised. A document (TBS) will provide information about the commands available to a user via the console, but there is also a "help" command. If you type "help xxx" the LTM will print out a short description of commands that contain "xxx" as part of the command or description. There are additional commands which are largely for the use of board initialization or software testing, but AMSAT expects that host users will not need these.

2.3.3 Software Telemetry and Commanding Overview

More details are provided below, but here we will provide an overview of how downlinked telemetry and uplinked commands relate to the LTM and the host satellite

- Host Telemetry – The LTM provides two forms of telemetry for the host called generic and opaque.
  - Generic telemetry means that the LTM software and the FoxTelem ground software 'understand' something about what the data mean and in addition that these data are generally sent to the FoxTelem "Health" tab and in health or safe mode. For example, the LTM gives the host the ability to specify number of temperature and voltage values as well as a number of single bit state values. Some of the temperature and voltage values are specifically intended to be used for solar panels, but the rest can be chosen by the host. The publicly-available FoxTelem will display the host-chosen values as (for example) Temp1, Temp2, Volt1, Volt2 etc. However, the hosts should be able to modify their own version of FoxTelem or provide a separate program for themselves and others to display telemetry using their own methods.
  - Opaque telemetry means that the LTM software and FoxTelem do not understand anything about the telemetry. It will be provided by the host satellite as CAN messages (see below for details), and transmitted such that FoxTelem can reproduce the CAN message, again with no understanding of what it means. Opaque telemetry operates slightly different depending on the mode that the LTM is operating in: Safe, Health, or Science.
    - Safe mode. In safe mode, the LTM attempts to reduce power use by sending less telemetry. Specifically, the LTM will transmit two frames of telemetry approximately every two minutes. Among other payloads, safe mode frames will include payloads consisting of opaque telemetry which the host designated as safe mode data. <TBS how many safe mode opaque payloads will be sent in a specified time>
    - Health mode. In health mode, the LTM continually sends health telemetry frames. Among other payloads health frames will include payloads consisting of opaque telemetry which the host designated to be either health mode or safe mode data. <TBS how many health mode opaque payloads will be sent in a given time.
    - Whole Orbit Data. In addition, a sample of most non-opaque data and whatever opaque data is specified by the host is saved temporarily as Whole Orbit Data. Approximately two orbits worth of data is stored and is transmitted repeatedly along with the real-time data. Thus, we will have access to data collected while the satellite is not close to a ground station.
    - Science mode. In science mode, the LTM continually sends frames which mostly consist of opaque telemetry that the host designated to be science data. A few payloads of health data will also be included. No science data is stored as Whole Orbit Data, so it should only be sent when near a ground station.

  In all cases, the host can choose to send this data to the LTM only during the mode it is intended for, or at any other time. The LTM will save the data in a queue and transmit it only during the specified mode. If the host sends more payloads than the LTM can save,

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the oldest payload for that mode is deleted without transmitting it. <It is TBS how many opaque CAN messages will be transmitted in each mode, and how many can be saved if they are not transmitted>

- Host Commanding – AMSAT provides a program called AMCOM to send commands to the satellite. As with telemetry, the LTM provides a number of commands that are generic, and also the ability to receive opaque commands and transfer them to the host satellite. Most generic commands are only used by the LTM, but mode change commands (Health, Safe, Science) are also passed to the host. Again, the hosts can provide their own software to input opaque commands into AMCOM to be sent to the satellite.

More software details:

2.3.4 Startup phases – Every time the LTM flight software starts, following a power cycle, following a reset, or following the first release from the launch vehicle, it passes through several phases of startup before reaching the main software. These phases are described below:

- Flash Loader (Startup Phase 1) – The first phase to execute is the flash loader. If the "Attached" digital signal is high, the flash loader waits (with a prompt of 'o') for a command from a terminal or the pyMicroLoader software on a PC to either continue to the next phase, or to load software from the attached PC into the LIHU's flash program memory. If pyMicroLoader supplies new code for the remaining startup phases and the main software, this code is written into the flash program memory and then executed. If there is just a command to continue, the existing code is executed. Note that the flash loader code is permanently installed in the processor's flash program storage and cannot be replaced except by having physical access to the LIHU.
- POST (several places in startup) – At several places during the remaining startup phases, power-on self-tests (POST) are run, including a CRC check on several sections of the flash program memory, a processor self-test, and a memory test. If any of these tests fail, the behavior is different depending on the state of the 'Attached' digital signal. If the 'Attached' signal is HIGH (for example, when on the ground in the lab) the CRC is recalculated and stored; other failure information is passed on to be displayed on the console. If 'Attached' is LOW for either case, the LIHU is power cycled in an attempt to fix what may be a deposited charge problem.
- Startup Completion: The after the above, the LTM pauses for a few seconds making an assortment of beeping noises. While the host is responsible for ensuring that there are no transmissions or deployments for a launcher-provided time following release from the rocket booster, and in addition is expected not to supply power to the LTM until the antennas are released, the LTM waits for a short additional time before doing any transmissions or listening for commands. Following this wait period, the LTM starts up in "Safe" mode, meaning that it transmits telemetry for about 10 seconds every 2 minutes with enough information for the host owner and AMSAT to gain some initial information before proceeding to enable additional features.

2.3.5 Major functions of the main software

The major functions of the LTM's flight software are to:

- Keep track of time (LTM Timekeeping),
- Receive and execute (or pass to the host platform) ground commands,
- Collect telemetry data from itself and the host platform and format it for downlinking,
- Implement a 1200 bps BPSK telemetry beacon,
- Implement operational modes that choose which telemetry data to send and when,
- Implement a Whole Orbit Data (WOD) collection scheme so telemetry can be collected at all times in orbit and then repeatedly downlinked so it is likely to be seen by some ground station(s), and
- Manage the LTM's amateur radio transponder based on ground commands and the operational mode being executed.

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2.3.6 Software Operational Modes - The LTM flight software, after startup is complete, has three basic operational modes: (1) Safe Mode, (2) Health Mode, and (3) Science Mode. In addition, the amateur radio transponder can be enabled by ground command, but will only be turned on in Health Mode. The operational mode is saved in non-volatile memory so that if the processor resets, it will re-enter the mode that it was previously in. Science Mode is slightly different in that it includes a timer so that it will return to Safe or Health mode when it times out, or if the processor should reset while it was active.

- Safe Mode: In Safe Mode, the intent is to attempt to reduce the power consumption of the satellite as much as possible. In particular, a CAN message is sent to the host so that it can reduce power, the amateur radio transponder is turned off, and the telemetry beacon is changed to be on for approximately 10 seconds (2 frames) every 2 minutes. The telemetry transmitted in Safe Mode is limited to health data from both the LTM and the host platform (including whole orbit health data). When first released from the launch vehicle, the LTM is in Safe Mode and will not change modes until commanded from the ground. In addition, the LTM has the capability of "auto-safe" which causes it to automatically enter and exit Safe Mode when a low voltage bus condition occurs. The host may also send a CAN message asking the LTM to switch to safe mode. It is recommended that this message be sent when the host senses that the battery is approaching the lowest voltage allowed.
- Health Mode: This is the "normal" mode in which the LTM operates. When in Health Mode, the LIHU continually sends telemetry, including health data, whole orbit data, and data of the host platform's choice. In Health Mode, it is possible to turn on the amateur radio transponder.
- Science Mode: When commanded into Science Mode, the LTM "remembers" its previous mode and sets a timer (specified within the Science Mode command). When the timer expires, the LTM returns to its previous mode. In Science Mode the transponder is turned off, and much more opaque telemetry is sent than in health mode. In addition, the host sends different CAN messages to be downlinked in Science mode. A few health frames will be sent during science mode to allow the ground station to keep track of any problems that might happe.

2.3.7 LTM Timekeeping - The LTM keeps track of time as a pair of numbers (epoch, seconds-in-epoch). The epoch number increments when the software initializes, i.e., following a reset. Thus, it is sometimes called a reset number. When the epoch increments, the seconds start at 0. The spacecraft time is guaranteed always to increment if the epoch number is considered more significant than seconds. It is included in every telemetry frame sent to the ground to indicate the time sent, in whole orbit data to indicate the time the data was collected, and in min/max data to indicate the last time a new minimum or maximum was observed.

The LTM is designed to expect an occasional reset due to failures that might occur from radiation, charged particle impacts, power problems or software bugs. Previous on-orbit experience has shown that a reset is common every few days and frequently happens near the South Atlantic Anomaly, implying that charged particles are a likely cause.

The LTM does not have a real-time clock that 'lives' across resets, nor does it have a GPS nor any other source of UTC.[1] However, on the ground, it is possible to correlate spacecraft time with UTC. This is accomplished by comparing the spacecraft time in a telemetry frame with the ground UTC when the telemetry was received, and using that to determine the UTC of the start of each epoch.

Optionally, the host may provide UTC time over CAN (received, for example, by a GPS unit). In this case, the spacecraft time (epoch, seconds) is still used as above, but an occasional telemetry frame that include the UTC time will be sent to allow automatic correlation of the spacecraft time with UTC. If the host does not provide time, the UTC telemetry data will be marked 'invalid'.

[1] UTC: Coordinated Universal Time—the basis for orbit calculations and most other time functions on earth.

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2.3.8 Telemetry

- Telemetry Format - Telemetry is downlinked as "frames" using BPSK (Binary Phase Shift Keying) at 1200 bits per second. All telemetry frames are the same size. The data consists of no more than 665 bytes, a 4-byte CRC check, and 3 32-byte Reed Solomon (255,223) check blocks for forward error correction (FEC). A frame requires roughly 6.5 seconds to send.

  A frame consists of a header followed by from 1 to 8 "payloads". The header includes the spacecraft time when the frame was assembled, the satellite ID number (assigned by AMSAT), the frame type, and the operational mode the LTM was in when the frame was assembled. Payloads consist of different kinds of data such as whole orbit data, host platform science data, LTM health data, and host platform health data. Each frame type has a specific set of payloads in a specific order. For example, frame type 1 might have LTM realtime health, LTM min values health, host platform opaque health data, LTM whole orbit data, host platform whole orbit data, etc. Type 2 might be the same but with LTM max values health. Type 3 might be 3 LTM whole orbit data payloads and 3 host platform whole orbit data payloads. The exact definitions of the data will be in a spreadsheet available from AMSAT. The frame types and frequency will depend on the mode of the spacecraft (Safe, Health, or Science) with details TBS.

- Payload Types[2]
  - **Health** – A health payload contains information about the operation of the spacecraft. Besides data from the LTM module (temperatures, radio power, voltages) it will also contain data gathered from the host I2c devices, the host-defined analog lines, host GPIO input lines and type 5 CAN data sent by the host. For data that has numerical values (generally those that come from A/D converters) the LTM also keeps track of the minimum and maximum values and downlinks those in "Min" and "Max" payloads.

  - **Health WOD** – A health WOD payload is nearly identical to a Health payload with these exceptions: 1) A WOD payload is collected and stored in a circular, non-volatile spacecraft buffer every 60 seconds. 2) The spacecraft time and mode when the payload was collected is also stored with the WOD payload. 3) Many (TBS) WOD payloads are downlinked per minute, so that each stored WOD payload will be sent multiple times before it is overwritten. Health and health WOD payloads are sent (at different rates) in Health and Safe mode.

  - **Host Opaque Health** – An opaque health payload consists of 83 bytes of data from the host spacecraft passed to the LTM over the CAN bus. Science data passed in a CAN message consists of the least significant 12 bits of the 29-bit CAN ID, 4 bits indicating the number of data bytes (0-8), and the specified number of data bytes. Thus, an opaque payload can consist of between 8 and 40 CAN messages (depending on how much data each one holds).

  - **Host WOD** – Similar to the Health WOD, the Host WOD contains opaque payloads that are saved in a non-volatile circular buffer. One significant difference is that the host can specify for a given CAN message whether it should be saved in WOD, downlinked immediately, or both.

[2] For this version of the document, the specifics of payload types should be considered examples. This section will be updated with more details in future versions.

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- **Long Science** – A long science payload fills an entire frame, and thus contains at most 656 bytes, or between 66 and 332 CAN messages. Long Science payloads are sent only in science mode.

- **Diagnostics** – This payload may include UTC time (if supplied).

- Telemetry Reception (FoxTelem) - FoxTelem is a program that runs on Windows, MacOS, or Linux, (including Raspberry Pi OS). It receives telemetry via either a software defined radio dongle (the RTL-SDR or the FunCube Dongle are supported) or direct sound card input from another radio. FoxTelem is designed to receive formats for many different satellites using AMSAT and the Partner's telemetry and to present the data in tabular or graphic form. The Partner's data will be labeled generically, e.g. "Temp 1" rather than "Battery 2 Temperature". FoxTelem also has the ability send this data to a central AMSAT server, to download data that was received around the world from that server, to output CSV tables of data, and to provide *"heat maps"* indicating various signal characteristics based on the area of the sky where the satellite was located when the signal was received. FoxTelem is open-source software (sources on GitHub, username AC2CZ) and is available for download from the AMSAT website. A minimum of version (TBD, greater than 1.12z6) will be required for the LTM.

2.3.9 Commanding

In accordance with FCC regulations, only a licensed amateur given permission by the satellite licensee is allowed to send commands to the satellite. Generally, this will mean a licensed Partner control operator, or an AMSAT control operator, depending on who the licensee is, and what permissions the licensee has given.
The following types of commands can be sent:

- Generic CAN message commands - A command can consist of a CAN message that is 'opaque' to the LTM and which is simply passed to the host platform.
- LTM Commands – These are commands common to most LTM-based satellites that are largely used to control the radio and operation modes, as well as override some automatic functions for testing and debugging.

Commands are generated by Windows or Linux software called AMCOM and provided by AMSAT. AMCOM output is via a sound card which can be connected to a standard amateur radio transmitter. AMCOM supports the Digirig Mobile (which includes a sound card and radio cables to send audio and transmitter-on (PTT) info to the chosen radio). AMCOM requires a username and password to operate. Each individual user is given a constellation of satellites and commands that they are allowed to send, as well as granted or not granted the capability to send commands from an external source via AMCOM. AMSAT provides the authentication file which specifies a user's name, his/her constellation of capabilities, and an initial password. The command uplink format includes forward error correction, whitening, interleaving, and secure authentication.

2.4 Signal Buses

2.4.1 Digital Signal Buses

- CAN Bus - The CAN bus will be the main data interface between the LTM and the Partner University's host platform. The CAN bus is a 5V bus on the LTM. However, due to the differential nature of the CAN bus, the host platform may use either a 5V or a 3.3V bus. The CAN bus will run at 1M bits/second. The CAN ID will be of the extended form using 29 bits. CANH and CANL lines are provided on pins H2/51 and H2/52 respectively of the PC-104 host platform interface (see Figure 5 for PC-104 pin details).

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- I2C Bus – The LTM provides an Inter-Integrated Circuit (I2C) bus which is slower and less capable than the CAN bus, but which may be used for direct connections to several I2C - equipped chips, specifically, the ADS7828 (12-bit, 8 input A-to-D converter chip) and the MAX31725 or 31726 (temp sensor). SCL and SDA I2C lines are available on the LTM PC-104 connector interface on pins H1/43 and H1/41 respectively (see Figure 5). The LTM will provide the required pull-up resistor on the I2C data lines. In addition, pin H1/29 is a "reset" line that is normally HIGH. When this line is lowered and raised, all I2C devices are expected to reset (if they have a reset capability) or power off and on.
- Serial Lines – The LTM provides a 57.6Kbps serial line to allow access to the LTM software for testing, checkout, and for loading new LTM software while the University satellite is on the ground. These lines are available on PC104 pins 39 and 40 for this purpose. The University should ensure that the serial lines are accessible via an umbilical after the satellite is built and integrated, and that the "Umbilical Attached" line is raised when the serial lines are in use. One way to do this is to provide a USB-to-Serial adapter chip and to use a USB mini, micro, or C connector as an umbilical connector. Then the USB power can be used to raise "Umbilical attached." In order to enable LIHU software uploading, it is CRITICAL that the "Umbilical Attached" (Pin 13 connector H2) signal line be held high when the USB umbilical is plugged in. **It is also CRITICAL that this line be held low in flight.**

2.4.2 Analog Telemetry Lines

There are four lines (named TLMn) on which the host platform may place analog voltages between 0 and 3.3V to be A to D converted to digital signals and sent to ground stations as telemetry data. If the host does not use these lines the values sent down via telemetry will be random and meaningless; AMSAT suggests that the host tie unused analog signals to ground to ensure the sent value is 0. These lines are on pins H1/20-23 of the PC-104 connector.

2.4.3 Discrete Digital Lines

- Umbilical Attached (input to LTM) - Pin H2/13 of the PC-104. The 'Attached' signal is an input to the LTM. The intent is that it will be raised when an umbilical connector is attached to the spacecraft. In addition, this signal is raised if a USB connection with power is plugged into the USB mini-B connector on the LTM board itself (see software section 2.5.4 for behavior).
- GPIOn (input or output to LTM) - There are four digital signal lines available on the PC-104 connector (pins H2/14-17); these are used for telemetered digital outputs from the host.

2.4.4 Antenna

The host is responsible for providing UHF and VHF antennas with a nominal impedance of 50 ohms. (Coax, etc.). The LTM provides a coax stub with SMA male connectors to connect to the host-supplied antennas.

2.5 LTM Power Requirements

The host platform's power system is responsible for generating power, maintaining battery health, and providing power to all CubeSat subsystems, including the LTM. Electrical power to the LTM shall be supplied via the LTM's V_{sys} lines. The V_{sys} voltage range should be 5.0V ±5%. Upon deployment from the launch vehicle's dispenser, it is the host's responsibility to withhold power from the LTM until it is safe for the LTM to transmit. The LTM requires that the host-supplied antennas be deployed before it receives power; in addition, the launch provider or integrator typically requires that nothing be deployed or transmitted until a specific time following release of the satellite from the launch vehicle or space station. This timing is the responsibility of the host.

The host platform design is responsible for the dispenser deployment separation switches and the Remove Before Flight (RBF) pin, and must ensure the LTM remains UNPOWERED after the host platform satellite is integrated, tested, and prepared for launch and before the satellite is released from the launch vehicle, with the following exception. If the 'Attached' signal is HIGH, and an umbilical is plugged into the

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satellite for battery charging, power **may** be supplied to the LTM. If the 'Attached' signal is HIGH, and an umbilical is plugged into the satellite for testing, power **must** be supplied to the LTM. Table 5 below shows the current design measurements of the power per LTM board.

PCB	ESTIMATES (from Eng Model Measurements)	COMMENTS
Improved Command Receiver (ICR)	200 mW	
Receiver/Transmitter (RX/TX)	2.6W (P_{OUT-PK} = 30 dBm)	With Average Power Tracking (APT); Power Dissipation will be dynamic, depending on Linear Transponder loading
LTM Internal Housekeeping Unit (LIHU)	100 mW	
TOTAL DC POWER INPUT	≈ 2.9 Watts (max.)	All power numbers are worst case maximum/peak values with all Communication functions active. Average power dissipation will depend on transmission duty cycle.

Table 5: LTM Power Requirements

2.6 Software and Data Connections and Protocols

2.6.1 CAN

- CAN Bus – The CAN bus will be the main data interface between the LTM and the host satellite. It will be used by the LTM to send commands received from the ground to the host platform, and to send telemetry information from the host platform to the ground via the LTM's telemetry capabilities. CAN will also be used to pass information between the LTM and the host platform. For example, the LTM will tell the host platform that it has switched to Science, Health, or Safe Mode thus will downlink different types and quantities of telemetry from the host platform. Similarly, the host platform can signal the LTM that the LTM should go into Safe Mode to reduce power consumption. The CAN bus may be used internally by the host platform or by the LTM using identifiers compatible with the list specified herein.
- Message Format – CAN messages will comprise a 29-bit ID, and a DLC specifying the number of data bytes from 0 to 8 data as required by the CAN specification. The format of the IDs required by the LTM is shown in Table 6 below.
- Validity bits—Because CAN messages are sent at times determined by the sender, CAN health telemetry message groups each set a "valid" bit. This bit is cleared after some time if no further messages are received. (The time can be specified by a console command before launch.) The bits are sent as part of the health telemetry. Thus FoxTelem users can tell if CAN health telemetry was received recently by the LTM.
- Specific Messages Between LTM and Host – See the tables below.
- FoxTelem Display: The default labels as supplied by AMSAT for telemetry values are shown. But these labels can be changed by the University Partner for their copy of FoxTelem.
- Messages Uplinked from the ground. Any CAN message can be sent from the ground to the host with the following restrictions: 1) ID bits 28:25 (priority) are not uplinked, and will be filled in by the LTM. They must be ignored by the host. 2) Bits 15:12, the destination field (see below) must be in the host's range (8 or above). Any number of data bytes between 0 and 8 may be sent.

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BIT NUMBERS	PURPOSE	Comments and Specific Values When Required
28:24 (5 bits)	Priority	If multiple messages are sent on the same bus simultaneously, the hardware prioritizes lower ID numbers. The host platform and LTM must accept messages with any priority and ignore the priority when interpreting the message. For opaque uplinked CAN messages, bit 24 may be used by the host for any purpose. Bits 28:25 will be set by the LTM software to ensure a high priority
23:20 (4 bits)	Source	The host platform must put a number 8 or above in this field if the destination is 3 or 0 (that is the LTM). Otherwise, the host platform may use this field as it wishes. The LTM will place 3 in this field when sending to the host platform. For opaque uplinked CAN messages these bits may be used for any purpose
19:16 (4 bits)	Type	The LTM will treat messages received from the host platform in accordance with their type as follows: 1,9: Telemetry to be downlinked as opaque CAN messages during safe mode or health mode. 2,10: Telemetry to be downlinked as opaque CAN messages during health mode only 3: Telemetry to be downlinked as opaque CAN messages as during science mode only. Note that only occasional opaque messages are downlinked in safe mode and health mode (more in health mode). In both cases, they are intermixed with non-opaque safe and health mode data. In science mode, the opaque messages are the majority of what is downlinked with an occasional frame of non-opaque health info. If the LTM runs out of science opaque data in science mode, it will insert health opaque data. 4: Passing status information (problem requiring a switch to Safe Mode for example) to coordinate actions between the host platform and the LTM. (see software section 2.1.3.7.) 5: Telemetry to be displayed on FoxTelem as health data and saved in health whole orbit data 8,9,10: Telemetry to be saved as Whole Orbit Data (WOD) (Note that adding 8 to telemetry types 1 and 2 means to send both in realtime AND saved as WOD. 8 stores in WOD only)

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		The LTM will use the following values when sending messages to the host platform: **4: Passing status information (such as operational mode, or spacecraft elapsed time) to coordinate actions between host platform and the LTM** **For an opaque uplinked command, this field may be used for any purpose.**
15:12 (4 bits)	**Destination**	**The LTM will only receive messages from the host platform which have these ID bits set to 3 or 0. The LTM will put 8 in this field when transferring uplinked non-opaque commands or status information to the host platform. The host platform must ignore messages that have this field set to less than 8. The LTM will ignore received messages with number 8 or greater and any such messages may be freely used within the host platform.** **For opaque uplinked commands, this field must be 8 or greater or the LTM will not send the command to the host.**
0:11 (12 bits)	**Message ID**	**For opaque CAN status sent from the host platform, this field is sent as part of the telemetry and is "opaque" to the LTM.** **For opaque uplinked commands sent to the host platform, this field may be used for any purpose.** **For status messages in either direction see Table 7.** **For health messages, see Table 8.**

Table 6: CAN Identifiers

Message ID (Bits 0:11)	Meaning	Data Units and Scale
0	Ignored	None
1	Enter Safe Mode	If LTM is source: Data 0=reason; 0=commanded, 1=low power If host is the source: TBD
2	Enter Health Mode	Data 0=reason; 0=commanded;1=power ok; 2=timeout of science
3	Enter Science Mode	Data 0=reason; 0=commanded; 1=Scheduled Time; Data 1=timeout (in minutes)

Table 7: CAN Status Messages (Type 4)

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Message ID (Bits 0:11)	Meaning	Data Units and Scale
8	Solar Panel Temperatures (displayed and labelled on FoxTelem health page)	Data 0=+X, Data 1=-X, Data 2,3=+-Y Data 4,5=+-Z Unit: Degrees C, Scale: 0-255 = -20C to +107.5C
9	Other Temperatures (by default displayed as Temp 0, Temp 1, to 8 on FoxTelem health page	Up to 8 temperatures in data 0-7 Unit: Degrees C Scale 0-255 = -20C to +107.5C
16	Solar Panel Voltages (by default displayed and labelled on FoxTelem health page)	Data 0=+X, Data 1=-X, Data 2,3=+-Y Data 4,5=+-Z Unit: Volts Scale: 0-255 = 0-10V
17	Other Voltages (by default displayed and labelled as Voltage 0, 1, etc on FoxTelem Health Page)	Up to 8 voltages in data 0-7 Unit: Volts Scale: 0-255 = 0-10V
24	Binary State Data 1 labelled as States 0-7)	8 bits in data 0 Unit: True/False
25	Binary State Data 2 (labelled as States 8-16)	8 bits in data 0 Unit: True/False
26	Binary State Data 3 (labelled as States 16-23)	8 bits in data 0 Unit: True/False
27	Binary State Data 4 (labelled as States 24-31)	8 bits in data 0 Unit: True/False
32	UTC Time (from GPS for example)	Data0=Year, Data1=Month (1-12), Data2=Day (1-31), Data3=Hour (0-23), Data 4=Minute (0-59), Data 5=Second (0-59)

Table 8: CAN Downlink Health Telemetry Messages (Type 5)

2.6.2 I2C Bus

- Protocol – The LTM is always the I2C master and thus the host platform cannot initiate any I2C transactions. The I2C bus will run in standard mode; speed is not guaranteed, but currently it is 10K bits/second. The LTM, when accessing the allowed I2C devices, will follow the formats specified by the datasheets of the respective devices.
- Devices and Addresses – The LTM will provide I2C support for a limited number of devices as listed in Table 9. The addresses for the available I2C devices are also listed in Table 9. Note that the host platform circuit design is responsible for choosing the correct addresses. Use of these I2c devices is optional. If they are unused, a value indicating that they are invalid will be send in telemetry. It is possible that other devices within the LTM will use the I2c bus but will always use different addresses than those specified in Table 9.

I2C ADDRESS	HOST PLATFORM SLAVE DEVICE	COMMENTS

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0x48-4B	**ADS7828**	**Programmed as single ended; raw data downlinked in telemetry. A maximum of 4 ADS7828s may be used by the host platform.**
0x4C-4F,54-57[3]	**MAX31725/6[4]**	**Temperature**

Table 9: I2C Devices and Addresses
(when not specified, max. number of devices TBD)

2.6.3 Umbilical

As described in section 2.3.1, the LTM provides a serial line for use in testing and software updating. Data for this line MUST be accessible via an umbilical when the satellite is closed up. One method is suggested in section 2.3.1. No matter how it is done, the "Umbilical Attached" line must be high when a terminal or PC is connected to the umbilical. This should be wired by the host to prevent antenna deployment and will be used by the LTM to disable the transmitter.

When used as a "console" the serial line should be connected to a terminal emulator. When used to load software, a PC (Linux, Windows, or Mac) program supplied by AMSAT can be used to load the LTM code. (See '*Attached*' line below for related information.

2.6.4 **Discrete Connections** – A discrete connection means a connection that is used for single purpose and has a single source and single destination. The LTM provides 9 discrete connections, described below.

- Analog Telemetry Lines – There are four lines called tlm1-tlm4 on which the host platform may place an analog voltage between 0 and 3.3V. These voltages will be digitized at 8 bits per value collected about every four seconds and sent to the ground by telemetry. The LTM samples these lines asynchronously, although not every sample will be sent by telemetry. It is the host platform's responsibility to ensure that these signals are meaningful at all times. For example, if this voltage varies and only the average is meaningful, the host platform must employ some sort of averaging circuitry. The host may choose not to use these lines, in which case AMSAT recommends that they be connected to ground.
- Digital Command Lines
  - *Umbilical Attached (input to LTM):* When high, 'Attached' inhibits the radio from transmitting and avoids some initial testing or timing for faster startup when "on the bench." AMSAT suggests that the host also use this line to avoid sending any signals to release deployables. The intent is that it would be raised when an umbilical connector is attached to the spacecraft. In addition, this signal is raised if a USB connection with power is plugged into the USB mini-B connector on the LIHU board itself. Another function of 'Attached' is that when it is high and the LTM is powered up or reset, the flash loader is started initially rather than the main flight software. This allows the user to load new AMSAT-supplied code for the LTM using the AMSAT supplied pyMicroloader software. If no code is to be loaded, a terminal command (the letter 'a') to the boot loader will start the flight code.
  - *GPIO1,2,3,4:* These signal lines are four digital inputs to the LTM. They are used as true/false signals from the host that will be telemetered to the ground. One example of a use the host might

[3] Note that the datasheet for the MAX31725 specifies the address as a full 8 bits with the least significant bit always 0. These addresses are listed in the more common 7-bit form. Thus what the ICD calls address 0x4C, the datasheet calls address 0x98, 4D=9A,54=A8, etc.
[4] Both MAX31725 and 31726 have these addresses available. The 31726 has an I2c reset line which can be connected to PC104 H1 pin 29.

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choose is to indicate that deployables have been released. (AMSAT highly recommends that this information be downlinked in some manner)

3. **ACRONYMS GLOSSARY**

ADAC	Attitude Determination & Control
ADC	Analog to Digital Converter
AMCOM	AMSAT Command (s/w pgm)
AMSAT	Radio Amateur Satellite Corporation
APT	Average Power Tracking
ASCII	American Std Code for Info Interchange
ATP	Acceptance Test Procedure
BPSK	Binary Phase Shift Keying
CAN	Controller Area Networking
CDR	Critical Design Review
CONOPS	Concept of Operations
CMD	Command
CRC	Cyclic Redundancy Check
CW	Continuous Wave (Morse Code)
dB	deci-Bel
dBc	dB's below Carrier
dBm	dB's relative to 1mW
DC	Direct Current
DLC	Data Length Code
EAR	Export Administration Regulations
EM	Engineering Model
EMC	ElectroMagnetic Compatibility
EMI	ElectroMagnetic Interference
EPS	Electrical Power System
FCC	Federal Communications Commission
FM	Flight Model
FoxTelem	Fox Telemetry (s/w pgm)
FR-4	Glass Reinforced Epoxy Laminate PCB
GND	Ground
GPIO	General Purpose Input - Output
GPS	Global Positioning System
I2C	Inter-Integrated Circuit
IARU	International Amateur Radio Union
IAW	In Accordance With
ICD	Interface Control Document
ICR	Improved Command Receiver
ID	Identification; IDentifier
IMD	Inter-Modulation Distortion
I/O	Input / Output
IT	Information Technology
ITAR	International Traffic in Arms Regulations
ITU	International Telecommunications Union
LIHU	Legacy Internal Housekeeping Unit
LO	Local Oscillator
LT	Linear Transponder
LTM	Linear Transponder Module
mA	milli-Ampere
MCU	MicroController Unit
MDS	Minimum Discernable Signal
MHz	Mega Hertz
MRR	Mission Readiness Review
mW	milli-Watt
NDA	Non-Disclosure Agreement
NF	Noise Figure
OBC	On Board Computer
OIP3	Output Intercept Point 3rd (Order)
PC	Personal Computer
PCB	Printed Circuit Board
PDR	Preliminary Design Review
POST	Power On Self Test

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ppm	parts per million	TBD	To Be Determined
QTH/QSH	AMSAT Fox CubeSat bus connectors	TLM	Telemetry
RBF	Remove Before Flight	UHF	Ultra High Frequency
RF	Radio Frequency	USB	Universal Serial Bus
RX/TX	Receiver/Transmitter (LTM card)	UTC	Coordinated Universal Time
SAT	Satellite; Saturated	VHF	Very High Frequency
SPI	Serial Peripheral Interface	WOD	Whole Orbit Data
SSB	Single Side Band	XCVR	Transceiver

4 REFERENCES/APPLICABLE DOCUMENTS

1) CubeSat Design Specification, Rev. 14.1, February 9, 2022, The CubeSat Program Cal Poly SLO
2) ITU Radio Regulations, Edition of 2012, available from http://www.itu.int/publ/R-REG-RR-2012/en
3) Fox-1E Linear Transponder Communications Performance Analysis (Preliminary), AMSAT, DRAFT v0.4, 11 October 2016

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